

Q1. Suppose all end hosts pump data into the network at a constant rate. Would you prefer packet-switching over circuit switching in this case?[3]

Answer: Yes/No? _____

Reason:

Q2. Suppose all the end hosts send data in periodically i.e. they do not constantly send data into the network.. Would you prefer packet-switching over circuit switching in this case? [3]

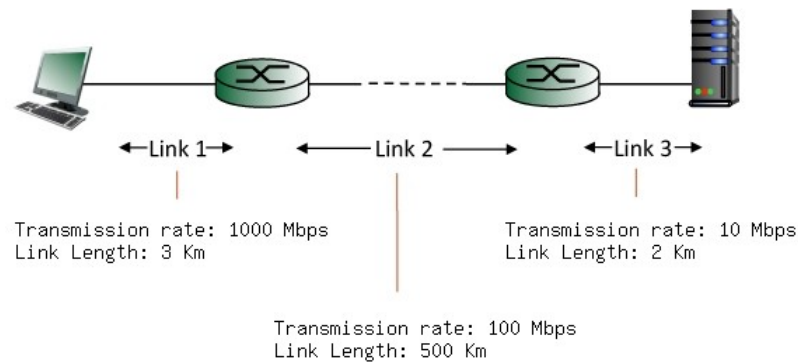
Answer: Yes/No? _____

Reason:

Q3: (a) Suppose N packets arrive simultaneously to a link at which no packets are currently being transmitted or queued. Each packet is of length L and the link has transmission rate R . What is the average queuing delay for the N packets? Show calculations to prove your answer.

(b) Now suppose that N such packets arrive to the link every LN/R seconds. What is the average queuing delay of a packet? Give reason to support your answer.

1. Consider the figure below, with three links, each with the specified transmission rate and link length.

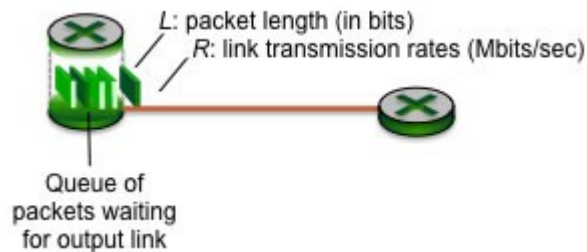


Find the end-to-end delay (including the transmission delays and propagation delays on each of the three links, but ignoring queueing delays and processing delays) from when the left host begins transmitting the first bit of a packet to the time when the last bit of that packet is received at the server at the right. The speed of light propagation delay on each link is 3×10^8 m/sec. Note that the transmission rates are in Mbps and the link distances are in Km. Assume a packet length of **16000** bits. Give your answer in milliseconds. [10+10+10=30 Marks]

Link 1 transmission delay = $L/R = 16000 \text{ bits} / 1000 \text{ Mbps} = 0.016000 \text{ msec.}$
Link 1 propagation delay = $d/s = 3 \text{ Km} / 3 \times 10^8 \text{ m/sec} = 0.010000 \text{ msec.}$
Link 2 transmission delay = $L/R = 16000 \text{ bits} / 100 \text{ Mbps} = 0.160000 \text{ msec.}$
Link 2 propagation delay = $d/s = 500 \text{ Km} / 3 \times 10^8 \text{ m/sec} = 1.666667 \text{ msec.}$
Link 3 transmission delay = $L/R = 16000 \text{ bits} / 10 \text{ Mbps} = 1.600000 \text{ msec.}$
Link 3 propagation delay = $d/s = 2 \text{ Km} / 3 \times 10^8 \text{ m/sec} = 0.006667 \text{ msec.}$

Thus, the total end-to-end delay is the sum of these six delays: 3.459333 msec.

2. Consider the figure below, in which a single router is transmitting packets, each of length L bits, over a single link with transmission rate R Mbps to another router at the other end of the link.



Suppose that the packet length is $L = 12000$ bits, and that the link transmission rate along the link to router on the right is $R = 100$ Mbps.

- (a) What is the transmission delay (the time needed to transmit all of a packet's bits into the link)?
- (b) what is the maximum number of packets per second that can be transmitted by the link?

[10+10=20 Marks]

- (a) The link transmission delay = $L/R = 12000 \text{ bits} / 100 \text{ Mbps} = 0.120000 \text{ msec}$.
- (b) The link can transmit 8333.333333 packets per second

1.1 What is the maximum achievable end-end throughput (in Mbps) for each of four client-to-server pairs, assuming that the middle link is fair-shared (i.e., divides its transmission rate equally among the four pairs)? Provide a brief explanation of your answer.

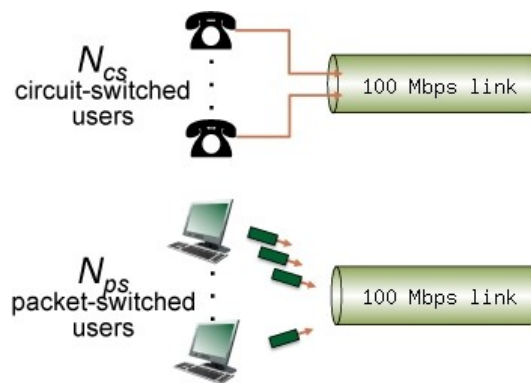
1.2 Which link is the bottleneck link for each session? and Why?

1.3 Assuming that the senders are sending at the maximum rate possible, what are the link utilizations for the sender links (R_S), client links (R_C), and the middle link (R)?

2. Consider the two scenarios below:

- A circuit-switching scenario in which N_{cs} users, each requiring a bandwidth of 25 Mbps, must share a link of capacity 100 Mbps.
- A packet-switching scenario with N_{ps} users sharing a 100 Mbps link, where each user again requires 25 Mbps when transmitting, but only needs to transmit 30 percent of the time.

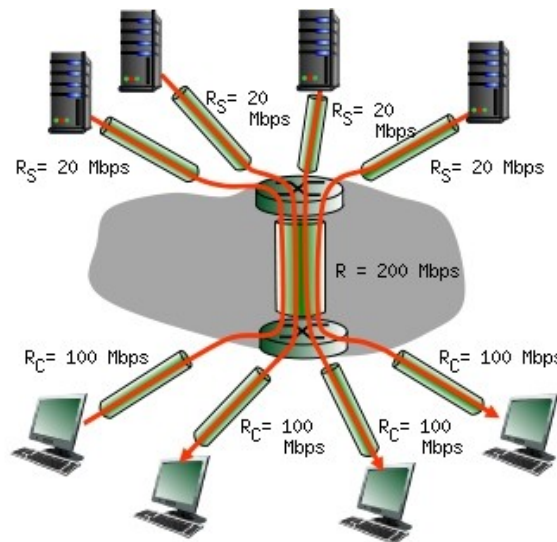
[10+10 = 20 Marks]



2.1 When circuit switching is used, what is the maximum number of circuit-switched users that can be supported? Explain your answer.

2.2 Now suppose that packet switching is used. Suppose there are 7 packet-switching users (i.e., $N_{ps} = 7$). Can this many users be supported under circuit-switching? Explain.

1. Consider the scenario shown below, with four different servers connected to four different clients over four three-hop paths. The four pairs share a common middle hop with a transmission capacity of $R = 200$ Mbps. The four links from the servers to the shared link have a transmission capacity of $R_S = 20$ Mbps. Each of the four links from the shared middle link to a client has a transmission capacity of $R_C = 100$ Mbps per second. [10+10+10=30 Marks]



1.1 What is the maximum achievable end-end throughput (in Mbps) for each of four client-to-server pairs, assuming that the middle link is fair-shared (i.e., divides its transmission rate equally among the four pairs)? Provide a brief explanation of your answer.

The maximum achievable end-end-throughput is 20 Mbps.

1.2 Which link is the bottleneck link for each session? and Why?

This is the transmission capacity of the first hop, which is the bottleneck link, since the first-hop transmission capacity of 20 Mbps is less than one quarter of the shared-link transmission capacity ($200/4 = 50$ Mbps) and less than the third-hop transmission capacity of 100 Mbps.

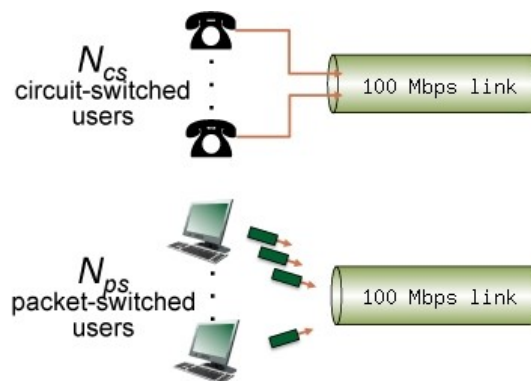
1.3 Assuming that the senders are sending at the maximum rate possible, what are the link utilizations for the sender links (R_S), client links (R_C), and the middle link (R)?

The utilization of sender links is 100% . The utilization of receiver links is 20% . The utilization of the middle link is 40% .

2. Consider the two scenarios below:

- A circuit-switching scenario in which N_{cs} users, each requiring a bandwidth of 25 Mbps, must share a link of capacity 100 Mbps.
- A packet-switching scenario with N_{ps} users sharing a 100 Mbps link, where each user again requires 25 Mbps when transmitting, but only needs to transmit 30 percent of the time.

[10+10 = 20 Marks]



2.1 When circuit switching is used, what is the maximum number of circuit-switched users that can be supported? Explain your answer.

When circuit switching is used, at most 4 circuit-switched users that can be supported. This is because each circuit-switched user must be allocated its 25 Mbps bandwidth, and there is 100 Mbps of link capacity that can be allocated.

2.2 Now suppose that packet switching is used. Suppose there are 7 packet-switching users (i.e., $N_{ps} = 7$). Can this many users be supported under circuit-switching? Explain.

No. Under circuit switching, the 7 users would each need to be allocated 25 Mbps, for an aggregate of 175 Mbps - more than the 100 Mbps of link capacity available.

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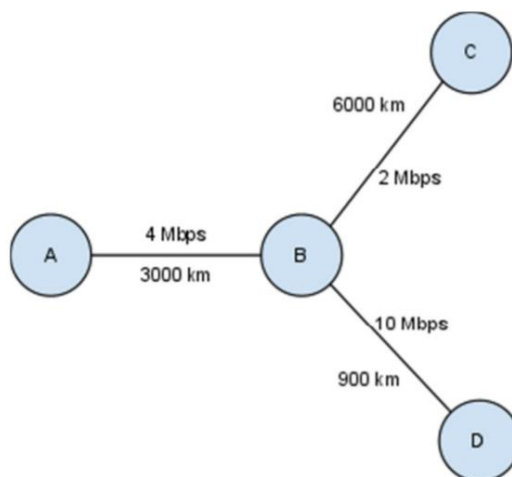
Course: Computer Networks
Program: BS(Computer Science)
Duration: 20 Minutes
Date: 11 Sep, 2019
Section: E

Course Code: CS307
Semester: Fall 2019
Total Marks: 20
Quiz: 1
Page(s): 1

Name _____ Roll No. _____

Question 1: [Marks 8]

Assume data travels through the links at the speed of light.



(a) What is the transmission delay if

- A sends a 500byte packet to B
- B sends a 500byte packet to D

(b) What is the propagation delay between

- A to B
- B to D

Question 2: [Marks 6]

A wants to send a 500byte packet to D through B. B is supposed to follow the store-and-forward model, that is, B will receive the whole packet from A and then start transmitting the packet to D.

- (a) What is the **end-to-end delay** seen by the packet?
- (b) What will be the throughput from A to D?

Question 3: [Mark 6]

- (a) If D starts sending 500 byte packets back-to-back to B, then how many packets will D have transmitted before B starts receiving the first packet sent by D?
- (b) What does this value have to do with the term “**bandwidth-delay product**”? (Extra Credit) [Marks 3]

- 1) (a) Transmission Delay = Size of Transfer / Link Bandwidth
A to B: Transmission Delay = $(500 / ((4/8) \times 10^6)) = \mathbf{0.001s \text{ or } 1ms}$
B to D: Transmission Delay = $(500 / (10/8 \times 10^6)) = \mathbf{0.0004s \text{ or } 0.4ms}$

(b) Propagation Delay = Distance of link / Speed of light
A to B:
Propagation Delay = $3000 / (3 \times 10^5) = \mathbf{10ms \text{ or } 0.01s}$
B to D: Propagation Delay = $900 / (3 \times 10^5) = \mathbf{3ms \text{ or } 0.003s}$

- 2) (a) End to end delay between A to D = (Delay between A to B) + (Delay between B to D)
Delay on a link = Transmission Delay + Propagation Delay
Therefore, Delay between A to B = $1 + 10 = 11ms$, Delay between B to D = $0.1 + 3 = 3.1ms$.
End to end delay between A to D = $11ms + 3.1ms = \mathbf{14.1ms \text{ or } 0.0144s}$

(b) Throughput = $\min\{A-B, B-D\} = \min\{4mbps, 10mbps\} = \mathbf{4mbps}$

- 3) (a) For D to B, the propagation delay is 3×10^{-3} , and the bandwidth is 10Mbps.
Therefore,
Bandwidth-Delay product is $(10/8 \times 10^6) \times (3 \times 10^{-3}) = 3750 \text{ bytes}$
This translates to $3750/500 = \mathbf{7.5 \text{ packets}}$

(b) This is similar to computing the volume of a pipe. The amount of data that will be “in flight” on a network link is the product of its bandwidth and the propagation delay.

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Course: Computer Networks
Program: BS(Computer Science)
Duration: 20 Minutes
Date: 11 Sep, 2019
Section: E (b)

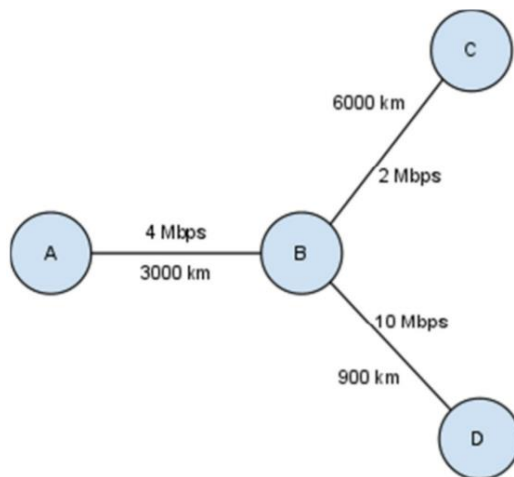
Course Code: CS307
Semester: Fall 2019
Total Marks: 20
Quiz: 1
Page(s): 1

Name _____

Roll No. _____

Question 1: [Marks 8]

Assume data travels through the links at the speed of light.



(a) What is the transmission delay if

- A sends a 700byte packet to B
- B sends a 700byte packet to C

(b) What is the propagation delay between

- A to B
- B to C

Question 2: [Marks 6]

A wants to send a 700byte packet to **C** through **B**. **B** is supposed to follow the store-and-forward model, that is, B will receive the whole packet from A and then start transmitting the packet to **C**.

- (a) What is the **end-to-end delay** seen by the packet?
(b) What will be the throughput from A to C?

Question 3: [Mark 6]

- (a) If **C** starts sending 700 byte packets back-to-back to **B**, then how many packets will C have transmitted before B starts receiving the first packet sent by C?
(b) What does this value have to do with the term "**bandwidth-delay product**"? (Extra Credit) [Marks 3]

- 1) (a) Transmission Delay = Size of Transfer / Link Bandwidth
 A to B: Transmission Delay = $(700 / ((4/8) \times 10^6)) = \mathbf{1.4ms \text{ or } 0.0014s}$
 B to C: Transmission Delay = $(700 / ((2/8) \times 10^6)) = \mathbf{2.8ms \text{ or } 0.0028s}$

 (b) Propagation Delay = Distance of link / Speed of light
 A to B:
 Propagation Delay = $3000 / (3 \times 10^5) = \mathbf{10ms \text{ or } 0.01s}$
 B to C: Propagation Delay = $6000 / (3 \times 10^5) = \mathbf{20ms \text{ or } 0.02s}$

- 2) (a) End to end delay between A to D = (Delay between A to B) + (Delay between B to D)
Delay on a link = Transmission Delay + Propagation Delay.
 Therefore,
 Delay between A to B = $1.4 + 10 = 11.4ms$
 Delay between B to C = $2.8 + 20 = 22.8ms$.
 End to end delay between A to D = $11.4ms + 22.8ms = 34.2$

 (b) Throughput = $\min\{A-B, B-C\} = \min\{4mbps, 2mbps\} = \mathbf{2Mbps}$

- 3) (a) For C to B, the propagation delay is 20×10^{-3} , and the bandwidth is 2Mbps.
 Therefore,
 Bandwidth-Delay product is $(2/8 \times 10^6) \times (20 \times 10^{-3}) = 5000 \text{ bytes}$
 This translates to $5000/700 = \mathbf{7.14packets}$

 (b) This is similar to computing the volume of a pipe. The amount of data that will be “in flight” on a network link is the product of its bandwidth and the propagation delay.

National University of Computer and Emerging Sciences, Lahore Campus
Quiz1 [BSCS: Section C] Fall 2022

Computer Networks (Code: CS3001)

Quiz Date: September 07, 2022

Total Marks: 12

Duration: 20 -Minutes

Name ----- Roll #----- Section -----

Instructions: Attempt all questions on this sheet. You can make use of rough sheet (do not attach to this sheet).

Q1: Describe the most popular wireless Internet access technologies today. Compare and contrast them.

(2 Marks) [CLO 2]

Q2: Solve the following questions:

(5+5 = 10 Marks) [CLO 4]

(A) Assume that you are downloading an MP3 file of 24 million bits (assume 1 million = 10^6 bits) from a server with one router between client host and server host. The transmission rate of the link from router to server i.e., R_s is 8 Mbps while transmission rate of your access link i.e., R_c is 6 Mbps. What will be the throughput? What is the required time to transfer the file considering that all delays except transmission delay are negligible? **(2+3 = 5 Marks)**

(B) Ten digital sources (channels) are multiplexed using TDM. If each input channel sends **5 kbps** (assume $1\text{ k} = 10^3$) and each output slot carries 8 bits from each digital source. Then for this link, find the (i) frame size in bits, (ii) frame rate, and (iii) bit duration? **(2+2+1 = 5 Marks)**

Start writing your Answers from here and then use backside of this sheet.

Answer Q1:

There are two popular wireless Internet access technologies today: **Wifi (802.11) and 3G and 4G wide-area wireless access networks**

Wifi (802.11) In a wireless LAN, wireless users transmit/receive packets to/from an base station (i.e., wireless access point) within a radius of few tens of meters. The base station is typically connected to the wired Internet and thus serves to connect wireless users to the wired network.

3G and 4G wide-area wireless access networks. In these systems, packets are transmitted over the same wireless infrastructure used for cellular telephony, with the base station thus being managed by a telecommunications provider. This provides wireless access to users within a radius of tens of kilometers of the base station

Solution Q2:

(A) Here, the only delay component is transmission delay.

Transmission rate of server = $R_s = 8 \text{ Mbps} = 8 * 10^6 \text{ bps}$

Transmission rate of access link = $R_c = 6 \text{ Mbps} = 6 * 10^6 \text{ bps}$

File size = $L = 24 \text{ million bits} = 24 * 10^6 \text{ bits}$

Throughput is equal to the transmission rate of bottleneck link i.e., $\min \{R_c, R_s\} = R_c = 6 * 10^6 \text{ bps}$

Having determined the throughput, delay i.e., time required to transfer the file = $L / \min \{R_c, R_s\}$

$$= L / R_c = 24 * 10^6 / 6 * 10^6 = 4 \text{ sec}$$

(B)

(i) Frame Size: # of digital sources * data from each source in a frame = $10 * 8 = 80 \text{ bits}$

(ii) Frame rate = bit rate / frame size

However, the bit rate of output link is equal to the bit rate of each digital source multiplied by the number of input sources i.e., $5 * 10^3 * 10 = 50000 \text{ bps}$

So Frame rate = $50000 / 80 = 625 \text{ frames/sec}$

(iii) bit duration = $1 / \text{bit rate} = 1 / 50000 = 0.00002 \text{ seconds}$

National University of Computer and Emerging Sciences, Lahore Campus
Quiz1 [BSCS: Section D] Fall 2022

Computer Networks (Code: CS3001)

Quiz Date: September 07, 2022

Total Marks: 12

Weight: 2%

Duration: 20 -Minutes

Name ----- Roll #----- Section -----

Instructions: Attempt all questions on this sheet. You can make use of rough sheet (do not attach to this sheet).

Q1: Differentiate between Circuit Switched and Packet Switched Networks. **(2 Marks)** **[CLO 2]**

Q2: Consider a packet of length 1000 bytes, which starts at source end system and travels over 10 links to destination end system. Nine packet switches connect these ten links. The transmission rate of all ten links is 2 Mbps. Suppose that propagation speed on all 10 links is 2×10^8 m/s. and each packet switch incurs a processing delay of 5 msec (assume zero processing delay at end systems). Moreover, suppose that the distance is the same i.e., 1000 km between all links (i.e., distance from source end system to packet switch 1, from packet switch 1 to packet switch 2, ..., and from packet switch 9 to destination end system is the same). Moreover, consider that no queuing delay exists, then what is the end-to-end delay for these values? **(3+3+2+2=10 Marks)** **[CLO 4]**

Start writing your Answers from here and then use backside of this sheet.

Answer Q1:

Circuit switched networks establishment connection before data transfer phase and then release connection after the data transfer phase. There is dedicated path between source and destination throughout the communication and data is delivered in order. The failure of a single switch is fatal for the communication.

While datagram approach of packet switched networks does not need to establish connection. Different packets can follow different path and hence can reach the destination out of order.

Solution Q4:

Length of packet = $L = 1000 \times 8 = 8000$ bits

Transmission rate of links: $R_1=R_2=R_3=R_4=...=R_{10}=2 \text{ Mbps} = 2 \times 10^6 \text{ bps}$

So Transmission delay of each link (L/R): $d_{trans1} = d_{trans2} = d_{trans3} = ... = d_{trans10} = 8000 / 2 \times 10^6 = 4 \text{ msec}$

Propagation speed of links: $s_1 = s_2 = s_3 = ... = s_{10} = 2 \times 10^8 \text{ m/s}$

Distance between links: $d_1 = d_2 = d_3 = ... = d_{10} = 1000 \text{ km}$

For one link = $d_{prop} = d/s = 1000 \times 10^3 / 2 \times 10^8 = 5 \text{ msec}$

So propagation delay of one link = 5 msec

Processing delay of each packet switch: $d_{\text{proc1}} = d_{\text{proc2}} = d_{\text{proc3}} = \dots = d_{\text{proc10}} = 5 \text{ msec}$

$d_{\text{end-end}} = 10 * d_{\text{trans}} + 10 * d_{\text{prop}} + 9 * d_{\text{proc}} = 10 \times 4 + 10 * 5 + 9 \times 5 = 40 + 50 + 45 = 135 \text{ msec}$

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Computer Science)	Semester:	Fall 2023
	Duration:	15 minutes	Total Marks:	15
	Quiz Date:	14-Sept-2023	Section	5J
	Exam Type:	Quiz 1 - Chapter 1	Page(s):	2

Student Name

Roll No.

Section:

Q1. Encircle the correct option:

[5 marks] [CLO 1]

- Which of the following is not a part of access network?
 - Edge Router
 - Server
 - Physical Media
 - Both (b) & (c)
- Which of the following delays are due to link properties?
 - Queueing & Transmission Delay
 - Propagation & Processing Delay
 - Propagation & Transmission Delay
 - Queueing & Processing Delay
- In which of the following a DSLAM is used
 - Edge Node
 - Cable Network
 - Optic Fiber
 - None of the above.

True/False:

- Transport Layer is implemented on a router. [T/F]
- Packet Switching uses Time Division Multiplexing. [T / F]

Q2. Assume that you are downloading an MP3 file of size 200 Mega bits ($1 \text{ Mega} = 10^6$). The transmission rate of server link is, i.e. R_s is 16 Mbps while transmission rate i.e., R_c of your access link is 4 Mbps. What is the required time to transfer the file considering that all delays except transmission delay are negligible? **[5 Marks] [CLO 1]**

Solution:

Q3.

(a) Which delay is directly proportional to the size of the packet? **[1 Mark] [CLO 1]**

(b) How long does a payload of 2 KB takes to propagate over a link of distance 1500 km, propagation speed 2.5×10^8 m/s, and transmission rate 4 Mbps? **[4 Marks] [CLO 1]**

Solution Quiz#1(5J)

MCQ's

1. b
2. c
3. d

T/F

1. F
2. F

Q2.

Transmission rate of server = $R_s = 16 \text{ Mbps} = 16 * 10^6 \text{ bps}$

Transmission rate of access link = $R_c = 4 \text{ Mbps} = 4 * 10^6 \text{ bps}$

File size = $L = 200 \text{ Mega bits} = 200 * 10^6 \text{ bits}$

Transmission rate of bottleneck link (i.e., R_c) will be used for the purpose of delay calculation because

$\min(R_s, R_c)$ is R_c .

Time required to transfer the file =

$$L / R = L / R_c = 200 * 10^6 / 4 * 10^6 = 50 \text{ seconds}$$

Q3.

(a) Transmission Delay

(b) Propagation delay = $d/s = 1500 * 1000 / 2.5 \times 10^8 = 6 \text{ ms}$ (The question only asks for propagation delay.)

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Computer Science)	Semester:	Fall 2023
	Duration:	15 minutes	Total Marks:	15
	Quiz Date:	14-Sept-2023	Section	5K
	Exam Type:	Quiz 1 - Chapter 1	Page(s):	2

Student Name

Roll No.

Section:

Q1. Encircle the correct option:

[5 marks] [CLO 1]

- Which of the following is not a host?
 - Server
 - Client
 - Router
 - Peer
- Which of the following delays are due to traffic mix and the switch/router internals?
 - Queueing & Transmission Delay
 - Propagation & Processing Delay
 - Propagation & Transmission Delay
 - Queueing & Processing Delay
- Frequency Division Multiplexing (FDM) is used by:
 - Packet Switching
 - Circuit Switching
 - Hybrid
 - None of the above

True/False:

- Packet Switching does not require congestion control and buffer management. [T / F]
- ISO/OSI reference model has 5 number of layers. [T / F]

Q2. Consider that the link capacity of a channel is 512 Kbps and round – trip time is 1000ms. What is the bandwidth delay product? **[5 Marks] [CLO 1]**

Solution:

Q3. In the transfer of file between server and client, if the transmission rates along the path is 10Mbps, 20Mbps, 30Mbps, 40Mbps. What will be the throughput? Explain your reason. **[5 Marks] [CLO 1]**

Solution:

Solution Quiz#1(5K)

MCQ's

1. c
2. d
3. b

T/F

1. F
2. F

Q2.

The bandwidth delay product = $512 \times 10^3 \text{ bits} \times \text{sec} \times 1000 \times 10^{-3} \text{ sec}$
= 512,000 bits = 64,000 bytes
= 62.5 KB

Q3.

The through put will be 10Mbps. The reason is that the through put is the minimum of all the transmission rate the occur inside its path. The minimum transmission rate is 10mbps so that will be the through put.

National University of Computer and Emerging Sciences, Lahore

Campus Quiz1 [BCS: Section 5A] Fall 2024

15

Computer Networks (Code: CS3001)

Quiz Date: September 03, 2024

Total Marks: 15

Duration: 20 -Minutes

Name _____

Roll # _____

Instructions: Attempt all questions on this sheet. You can make use of rough sheet (do not attach to this sheet).

Q1: Identify and encircle the correct option(s). (3 Marks) [CLO 1]

1- For over a hundred years, _____ has been used by telephone networks.

- (a) Twisted pair copper wire (b) Coaxial cable
(c) Fiber optic cable (d) None of the given options

2- The two main types of switching networks are:

- (a) Routers and Switches (b) Circuit-switched and Packet-switched
(c) Wi-Fi and Ethernet (d) None of these

3- Server is part of the

- (a) Edge Network (b) Core Network (c) Both Edge & Core Network (d) None of these

Q2: The first column of the table shows the function of a layer in the Internet protocol stack. You are required to write the correct name of the layer in second column that matches with the function provided in the first column. (5 Marks) [CLO 1]

Function of a layer	Layer Name
Protocols that are part of a distributed network application.	Application
Transfer of data between one process and another process (typically, on different hosts).	Transport
Delivery of datagrams from a source host to a destination host.	Network
Transfer of data between neighboring network devices.	Data-Link
Transfer of a bit into and out of a transmission media.	Physical

Q3: Consider a packet of length 3000 bytes, which starts at source end system and travels over four links to destination end system. Three packet switches connect these four links. Suppose that propagation speed on all four links is 2×10^8 m/s. The transmission rate of all four links is 2 Mbps and each packet switch incurs a processing delay of 5 msec. Moreover, suppose that the distance from source end system to packet switch 1 is 5,000 km, the distance from packet switch 1 to packet switch 2 is 4,000 km, the distance from packet switch 2 to packet switch 3 is 3,000 km, and the distance from packet switch 3 to destination end system is 1,000 km. What is the end-to-end delay for these values assuming no queuing delay exist? (7 Marks) [CLO 1]

Write your Answer for Q3 on backside of this sheet.

$L = 3000$ bytes; prop speed = 2×10^8 m/s; transmission rate = 2 Mbps
processing delay = 5 msec; $d_{\text{queue}} = 0$

Diagram: S — 5000 km — (1) — 4000 km — (2) — 3000 km — (3) — 1000 km — D

End-to-end delay = $d_{\text{trans}} + d_{\text{proc}} + d_{\text{prop}}$

$d_{\text{trans}} = N(L/R) = 4 \left(\frac{3000 \times 8}{2 \times 10^6} \right) = 0.048$

$d_{\text{proc}} = 3 \times (5 \times 10^{-3}) = 0.015$

$d_{\text{prop}} = \frac{(5000 + 4000 + 3000 + 1000) \times 10^3}{2 \times 10^8} = 0.0655$

Total delay = $0.048 + 0.015 + 0.0655 = 0.1285$ sec

Do with Pen

$$d_{\text{end-to-end}} = d_{\text{trans}} + d_{\text{prop}} + d_{\text{proc}}$$

$$= 4 \left(\frac{3000 \times 8}{2 \times 10^6} \right) + \left(\frac{(5000 + 4000 + 3000 + 1000) \times 1000}{2 \times 10^8} \right) + (3 \times 5 \times 10^{-3})$$

$$= 0.048 + 0.065 + 0.015$$

$$= 0.128 \text{ s}$$

$$= \boxed{128 \text{ ms}}$$



7

SOLUTION QUIZ-01 BCS-5A

QUESTION-01

1. b
2. c
3. d

TRUE/FALSE

- i. False
- ii. True

QUESTION-02

Part (1)

We have 3 links between Client and Server: (Transmission Delay: L/R , Propagation Delay: d/s)

For Link 1, the transmission delay = $(8 * 1500)/(5 \times 10^6) = 2.4\text{ms}$.

For Link 1, the propagation delay = $(160000)/(2.5 \times 10^8) = 0.64\text{ms}$.

For Link 2, the transmission delay = $(8 * 1500)/(10 \times 10^6) = 1.2\text{ms}$.

For Link 2, the propagation delay = $(250000)/(2.5 \times 10^8) = 1\text{ms}$.

For Link 3, the transmission delay = $(8 * 1500)/(50 \times 10^6) = 0.24\text{ms}$.

For Link 3, the propagation delay = $(100000)/(2.5 \times 10^8) = 0.4\text{ms}$.

Total end to end delay will be Transmission + Propagation + Processing delays.

$= (2.4 + 1.2 + 0.24) + (0.64 + 1 + 0.4) + (0.5 + 0.5) = 6.88 \text{ ms}$

Part (2)

- $5/2 = 2.5$. The floor of 2.5 is 2. So, 2 users can be maintained simultaneously.
- $P = 0.25 \times (0.75)^7 = 0.03337097167 = 0.033$ or 3.33 percent.
- $P = 56 \times 0.25^3 \times 0.75^5 = 0.2076$ or 20.7 percent

SOLUTION QUIZ-01 BCS-5C

QUESTION-01

1. a
2. b
3. b

TRUE/FALSE

- i. False
- ii. False

QUESTION-02

Part (1)

We have 3 links between A and B: (Transmission Delay: L/R , Propagation Delay: d/s)

For $A \rightarrow S1$, the transmission delay = $(8 * 1500)/(2 \times 10^6) = 0.006s = 6ms$.

For $A \rightarrow S1$, the propagation delay = $(3000000)/(3 \times 10^6) = 1s = 1000ms$.

For $S1 \rightarrow S2$, the transmission delay = $(8 * 1500)/(3 \times 10^6) = 0.006s = 4ms$.

For $S1 \rightarrow S2$, the propagation delay = $(4500000)/(3 \times 10^6) = 1.5s = 1500ms$.

For $S2 \rightarrow B$, the transmission delay = $(8 * 1500)/(3.5 \times 10^6) = 3.43ms$.

For $S2 \rightarrow B$, the propagation delay = $(1500000)/(3 \times 10^6) = 0.5s = 500ms$.

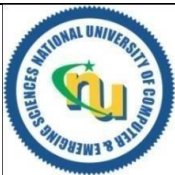
Total end to end delay will be Transmission + Propagation + Processing delays.

$= (6 + 4 + 3.43) + (1000 + 1500 + 500) + (2 + 1) = 3016.43 \text{ ms} = 3.016 \text{ sec}$

Part (2)

- $B \rightarrow S2$ is the bottleneck link (as it has the lowest transmission capacity, i.e. 3.5 Mbps).
- The bottleneck link in B to D is $S3 \rightarrow D$, with a capacity of 2 Mbps. If traffic is 1Mbps then utilization is $1/2$ which is 0.5 or 50%.

**National University of Computer and Emerging Sciences, Lahore
Campus**

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Computer Science)	Semester:	Fall 2025
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	04-September-2025	Section	5A , 5C
	Exam Type:	Quiz 1 - Chapter 1	Page(s):	2

Student Name

Roll No.

Section:

Q1. Encircle the correct option:

[5 marks] [CLO 1]

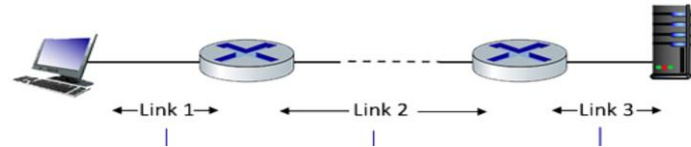
1. In a packet-switched network, each user transmits with probability 0.5 in a given time slot. What is the probability that a given user does not transmit?
a) 0.25
b) 0.50
c) 0.75
d) 0.98
2. Calculate the Bandwidth-Delay Product if the data rate is 100 Mbps and the round trip delay time is 10 milliseconds.
a) 100 Mbit
b) 1 Gbit
c) 1 Mbit
d) 10 Gbit
3. Which of the following delays is not a component of nodal delay?
a) Transmission delay
b) Propagation delay
c) Processing delay
d) Switching delay

True/False:

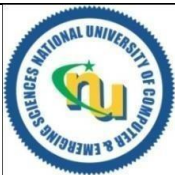
1. Internet Exchange Points (IXPs) are primarily used by end-users to directly connect their personal devices to the Internet. [T / F]
2. In packet switching, each packet may take a different route through the network before reaching the destination. [T / F]

Q2: Consider the following Network with 3 Links. You have to send a Packet of 1500 Bytes from Sender to Receiver. Propagation speed is 2.5×10^8 m/s. Link 3 is 100 km with transmission rate of 50 Mbps. Link 2 is 250 km long with rate 10 Mbps and Link 1 is 160 km long with rate of 5 Mbps. Processing delay is 0.5 ms for both switches (each). **[6 + 4 Marks] [CLO 1]**

1. Calculate the Transmission and Propagation delay for all links along with the End-to-end Delay (assume no queuing delay).
2. If circuit switching is used on the link 1 (5 Mbps) and each user transmits independently in a given time slot with probability $p = 0.25$, what is the maximum number of users that can be supported simultaneously, if each requires 2 Mbps? What is the probability that a given user is transmitting while all the other users are not transmitting? What is the probability that any 3 of the 8 users are transmitting simultaneously, and the remaining are not transmitting?



**National University of Computer and Emerging Sciences, Lahore
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	Program:	BS (Computer Science)	Semester:	Fall 2025
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	04-September-2025	Section	5A , 5C
	Exam Type:	Quiz 1 - Chapter 1	Page(s):	2

Student Name

Roll No.

Section:

Q1. Encircle the correct option:

[5 marks] [CLO 1]

1. A _____ is formed at the transport layer by encapsulating an application layer message with transport layer headers.
 - a) Segment
 - b) Datagram
 - c) Message
 - d) Frame
2. Which of the following best describes a Computer Network:
 - a) A collection of operating systems running simultaneously.
 - b) A group of interconnected devices that can exchange data
 - c) A single computer with multiple CPUs
 - d) A distributed database for storing web pages
3. In a circuit switched network, a link has 200 Mbps capacity. Each user requires 27 Mbps when active. What is the maximum number of users that can be supported simultaneously?
 - a) 5
 - b) 7
 - c) 10
 - d) 13

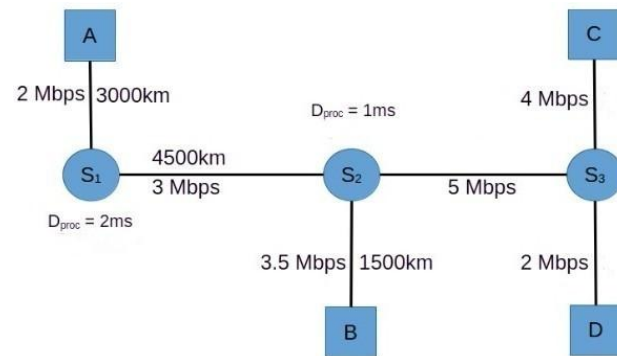
True/False:

1. TCP/IP Service Model consists of 7 layers. [T / F]
2. Transmission delay for equal sized packets for a given link is variable. [T / F]

Q2: Consider the following Network with Hosts A, B, C and D and Switches S₁, S₂ and S₃. Processing Delay for each Switch is given with D_{proc}. **[8 + 2 Marks] [CLO 1]**

1. Calculate the Transmission and Propagation delay for all links (A to B) along with the End-to-end Delay (one way) if a packet of size 1500 bytes is sent from A to B. Assume, no Queuing Delay and Assume propagation speed to be 3×10^6 m/s.

2. Which is the bottleneck link from B to C? What is the Link Utilization of B to D if traffic is 1Mbps?



SOLUTION QUIZ-01 BCS-6A

QUESTION-01

MCQs

- 1) A
- 2) A
- 3) C

TRUE/FALSE

1. False
2. True

QUESTION-02

(a) Transmission Delay = Size of Transfer / Link Bandwidth
 $= (500 \times 8) / (10 \times 10^6) = 4 \times 10^{-4} \text{ secs}$

(b) i. Ignore propagation, queuing, and processing delays

First Packet from Source Host to First Switch:

transmission Delay = $(250 \times 8) / (10 \times 10^6) = 2 \times 10^{-4} \text{ secs}$

Time at which 1st packet is received at the destination host = $2 \times 10^{-4} \times 2 = 4 \times 10^{-4} \text{ secs}$

After $(2 \times 10^{-4}) \text{ secs}$, 2nd Packet is Received at destination = $(4 \times 10^{-4}) + (2 \times 10^{-4}) = 6 \times 10^{-4} \text{ secs}$

ii. Throughput = $\min\{A-B, B-C\} = \min\{10\text{Mbps}, 2\text{Mbps}\} = \mathbf{2\text{Mbps}}$

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Computer Science)	Semester:	Spring 2024
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	13-Feb-2024	Section	6A
	Exam Type:	Quiz 1 - Chapter 1	Page(s):	2

Student Name

Roll No.

Section:

Q1. Encircle the correct option:

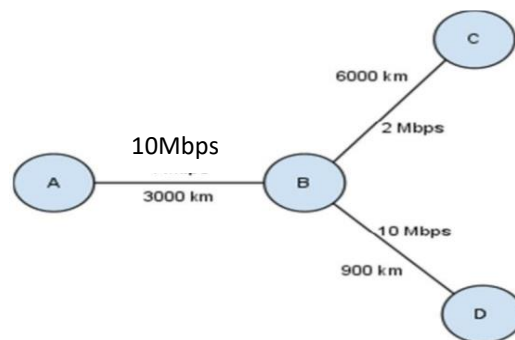
[5 marks] [CLO 1]

1. If the size of packet is increased the transmission delay _____ and if the bandwidth is increased transmission delay _____
 - a) increases, decreases
 - b) increases, increases
 - c) decreases, increases
 - d) decreases, decreases
2. Server is a part of
 - a) Network edge
 - b) Core network
 - c) both
 - d) none
3. Calculate the Bandwidth-Delay Product if the data rate is 100 Mbps and the round trip delay time is 10 milliseconds.
 - a) 100 Mbit
 - b) 1 Gbit
 - c) 1 Mbit
 - d) 10 Gbit

True/False:

1. OSI/ISO and TCP/IP Service Model consists of 5 and 7 layers respectively. [T / F]
2. If the traffic intensity is greater than 1 it means the average rate at which bits arrive at the queue exceeds the rate at which the bits can be transmitted from the queue. [T / F]

Question 2: Figure illustrates the end-to-end transport of a message with and without message segmentation. Ignore propagation, queuing, and processing delays. [10] [CLO 1]



(a) What is the transmission delay if A sends a 500 byte packet to B [3 Marks]

(b) Now suppose that the message is segmented into 2 packets, with each packet being 250-byte long. **A** wants to send message to **D** through **B**. **B** is supposed to follow the store-and forward model, that is, B will receive the whole packet from A and then start transmitting the packet to **D**.

- i. How long does it take to move the first packet from source host to the first switch? When the first packet is being sent from the first switch to the second switch, the second packet is being sent from the source host to the first switch. After how long is the 2nd packet received at the destination? [5 Marks]

- ii. What will be the throughput from A to C? [2 Marks]

SOLUTION QUIZ-01 BSE-6A

QUESTION-01

- 1) A
- 2) A
- 3) B

TRUE/FALSE

- i. False
- ii. True

QUESTION-02

Link 1 transmission delay = $L/R = 16000 \text{ bits} / 1000 \text{ Mbps} = 0.016000 \text{ msec}$.

Link 1 propagation delay = $d/s = 3 \text{ Km} / 3 \times 10^8 \text{ m/sec} = 0.010000 \text{ msec}$.

Link 2 transmission delay = $L/R = 16000 \text{ bits} / 100 \text{ Mbps} = 0.160000 \text{ msec}$.

Link 2 propagation delay = $d/s = 500 \text{ Km} / 3 \times 10^8 \text{ m/sec} = 1.666667 \text{ msec}$.

Link 3 transmission delay = $L/R = 16000 \text{ bits} / 10 \text{ Mbps} = 1.600000 \text{ msec}$.

Link 3 propagation delay = $d/s = 2 \text{ Km} / 3 \times 10^8 \text{ m/sec} = 0.006667 \text{ msec}$.

Thus, the total end-to-end delay is the sum of these six delays: 3.459333 msec.

Throughput is 10Mbps. (Throughput is the minimum of all the transmission rates that occur along the path from client to server).

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Software Engineering)	Semester:	Spring 2024
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	13-Feb-2024	Section	BSE-6A
	Exam Type:	Quiz 1 - Chapter 1	Page(s):	2

Student Name

Roll No.

Section:

Q1. Encircle the correct option:

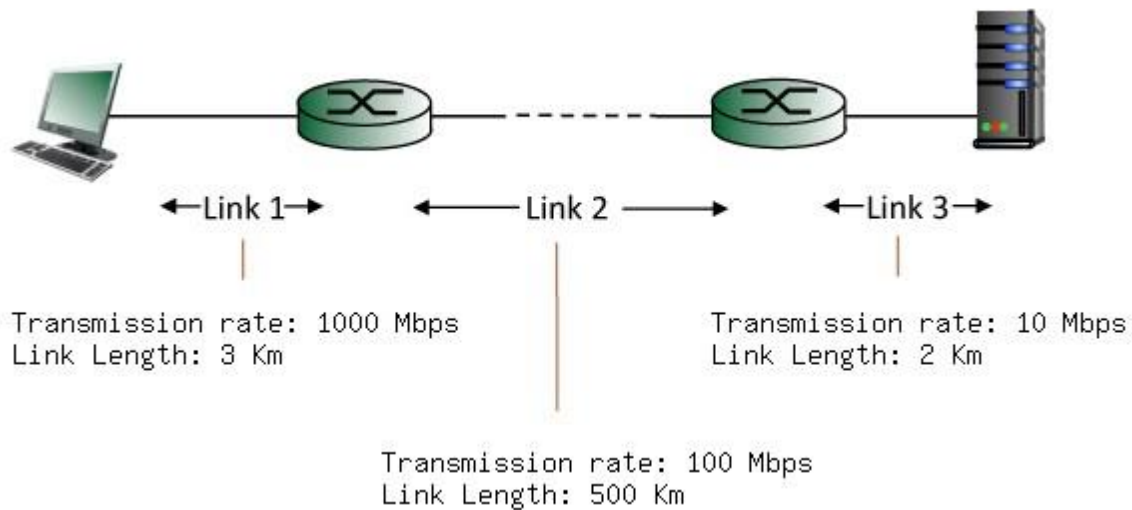
[5 marks] [CLO 1]

1. A network has a propagation speed of 2.5×10^8 meters per second. The distance between two nodes is 500 kilometers. Calculate the propagation delay
 - a) 2 milliseconds
 - b) 4 milliseconds
 - c) 6 milliseconds
 - d) 8 milliseconds
2. If the size of packet is increased the traffic intensity _____ and if the bandwidth is increased traffic intensity ____
 - a) increases, decreases
 - b) increases, increases
 - c) decreases, increases
 - d) decreases, decreases
3. When using Time Division Multiplexing (TDM), what happens during the time slots allocated to different channels?
 - a) All channels transmit data simultaneously.
 - b) Each channel takes turns transmitting data.
 - c) Channels transmit data on separate frequencies.
 - d) Channels transmit data in parallel.

True/False:

- i. OSI/ISO and TCP/IP Service Model consists of 5 and 7 layers respectively. [T / F]
- ii. In a store-and-forward network, the entire message must be received and stored at an intermediate node before it is forwarded to the next destination. [T / F]

Q2. Consider the figure below, with three links, each with the specified transmission rate and link length. **[8+1+1=10 marks] [CLO 1]**



- i. Find the **end-to-end delay** (including the transmission delays and propagation delays on each of the three links, but ignoring queueing delays and processing delays) from when the left host begins transmitting the first bit of a packet to the time when the last bit of that packet is received at the server at the right. The speed of light propagation delay on each link is 3×10^8 m/sec. Note that the **transmission rates are in Mbps** and the **link distances are in Km**. Assume a **packet length of 16000 bits**. Give your **answer in milliseconds**.

- ii. Also give the **throughput** between client and server.

SOLUTION QUIZ-01 BSE-6A

QUESTION-01

1. **Answer:** d) Access Delay

- *Explanation:* Access delay is not part of the four main delays (processing, transmission, propagation, and queuing delays).

2. **Answer:** a) Delay caused by signal transmission through a medium

- *Explanation:* Propagation delay is the time it takes for the signal to travel through a medium (e.g., fiber optic or copper wire).

3. **Answer:** a) DSLAM

- *Explanation:* DSLAM (Digital Subscriber Line Access Multiplexer) aggregates multiple DSL connections.

TRUE/FALSE

- False
- False

QUESTION-02

L = 1500bytes
S = 2.5×10^8
R = 2 Mbps

D trans = L / R 1500 bytes/ 2Mbps = 12,000bits / 2,000,000 = 0.006
secs = 6msec

Link 1 prop = D/S 5000km / 2.5×10^8 m/s = 5,000,000m / 2.5×10^8 =
20 msec

Link 2 prop = 4000km / 2.5×10^8 = 4,000,000 m / 2.5×10^8 = 16 msec

Link 3 prop = 1000km / 2.5×10^8 = 1,000,000 / 2.5×10^8 = 4msec

End to end = 6 + 20 + 3 + 6 + 16 + 3 + 6 + 4 = 64 msec

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (SE)	Semester:	Spring 2025
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	06-February-2025	Section	6A
	Exam Type:	Quiz 1 - Chapter 1	Page(s):	2

Student Name

Roll No.

Section:

Q1. Encircle the correct option:

[5 marks] [CLO 1]

Which of the following is **not a key delay** in the Internet's end-to-end communication?

- a) Propagation Delay
- b) Processing Delay
- c) Transmission Delay
- d) Access Delay

Which of the following best describes **propagation delay**?

- a) Delay caused by signal transmission through a medium
- b) Time taken by a router to process a packet
- c) Delay caused by queuing at a router
- d) Time taken to divide packets for transmission

Which device aggregates DSL connections at an ISP?

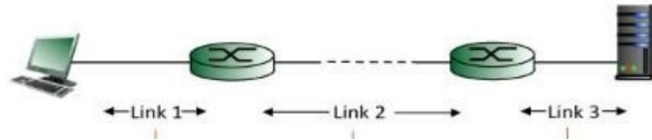
- a) DSLAM
- b) Router
- c) Modem
- d) Optical Node

True/False:

- i. Circuit switching is more efficient for large-scale networks than packet switching. [T / F]
- ii. The physical layer is responsible for managing application-level protocols. [T / F]

Question 2:**[Marks 10] [CLO 1]**

Consider a packet of length L that begins at end system A and travels over three links to a destination end system. These three links are connected by two packet switches. Suppose now the packet is 1,500 bytes, the propagation speed on all three links is $2.5 \times 10^8 \text{ m/s}$, the transmission rate of all the three links is 2 Mbps, the packet switch processing delay is 3 msec, the length of the first link is 5,000 km, the length of the second link is 4,000 km, and the length of the last link is 1,000 km. For these values, what is the end-to-end delay?



SOLUTION QUIZ-01 BSE-6C

QUESTION-01

1. **Answer:** c) Mobile Phone

- *Explanation:* End systems are devices like phones, computers, or IoT devices that directly communicate over a network. Routers and DSLAM are part of the infrastructure.

2. **Answer:** b) Queuing Delay

1. *Explanation:* Queuing delay occurs when packets wait in a router's queue due to congestion.

3. **Answer:** b) Resources are shared on demand.

- *Explanation:* In packet switching, resources like bandwidth and buffers are dynamically shared among all users.

TRUE/FALSE

- i. False
- ii. True

QUESTION-02

Segment 1:

Transmission Delay

The end system, ES1, pushes the 1,500 Byte packet onto the wire.

$$\frac{L}{R} = \frac{1500 \text{ B}}{2\text{Mbps}} = \mathbf{6 \text{ ms}}$$

Propagation Delay

Once the packet hits the wire, the propagation delay is how long it takes the bit to be sent over the wire to R1.

$$\frac{D}{S} = \frac{5000 \text{ km}}{2.5 * 10^8 \text{ ms}} = \mathbf{20 \text{ ms}}$$

Segment 2:

Transmission Delay

It's the same **6 ms** since the packet size is still 1,500Bytes and the transmission rate is still 2Mbps

Propagation Delay

The distance between R1 and R2 is now 4,000 km so we'll see a decrease in propagation delay to 16 ms.

$$\frac{D}{S} = \frac{4000 \text{ km}}{2.5 * 10^8 \text{ ms}} = \mathbf{16 \text{ ms}}$$

Segment 3:

Transmission Delay

Again, no change in L or R so it's still **6 ms**.

Propagation Delay

The distance from R2 to ES2 is 1,000km so again we'll see a lower propagation delay,

$$\frac{D}{S} = \frac{1000 \text{ km}}{2.5 * 10^8 \text{ ms}} = \mathbf{4 \text{ ms}}$$

Router Processing Delay

Just for the sake of completion, let's say the router vendor told us the router processing delay is 5 ms. So any packet going through R1 or R2 will automatically incur a 5 ms delay for packet processing.

Overall End-to-End Delay

- Segment 1 delay = **26 ms**
- Segment 2 delay = **22 ms**
- Segment 3 delay = **10 ms**
- Router Processing = **10 ms**

Total End-to-End Delay = 68 ms

**National University of Computer and Emerging Sciences, Lahore
Campus**

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (SE)	Semester:	Spring 2025
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	06-February-2025	Section	6C
	Exam Type:	Quiz 1 - Chapter 1	Page(s):	2

Student Name

Roll No.

Section:

Q1. Encircle the correct option:

[5 marks] [CLO 1]

Which of the following is an **end system** in a network?

- a) Router
- b) Link Layer Switch
- c) Mobile Phone
- d) DSLAM

Which delay is caused by **buffering and queuing** of packets at a router?

- a) Propagation Delay
- b) Queuing Delay
- c) Transmission Delay
- d) Processing Delay

In a **packet-switched network**, how are resources allocated for communication?

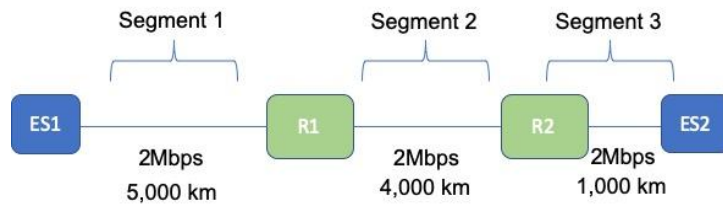
- a) Resources are pre-allocated along a dedicated path.
- b) Resources are shared on demand.
- c) All resources are allocated to the receiver.
- d) Resources are not allocated.

True/False:

- i. **Propagation delay** depends on the size of the packet. (T/F)
- ii. **Circuit switching** provides guaranteed bandwidth for a connection. (T/F)

Question 2:**[Marks 10] [CLO 1]**

A company has established a network connection between End System 1 (ES1) and End System 2 (ES2), which consists of three segments connected through two routers (R1 and R2). The first segment between ES1 and R1 spans 5000 km, the second segment between R1 and R2 spans 4000 km, and the third segment between R2 and ES2 spans 1000 km. Each link operates at a data rate of 2 Mbps. The packet size being transmitted is 1500 bytes and the propagation speed is 2.5×10^8 m/s. Additionally, both routers (R1 and R2) incur a processing delay of 5 milliseconds each. Calculate the transmission delay & propagation delay for each segment (Segment 1, Segment 2 and Segment 3) and also calculate the total end-to-end delay from ES1 to ES2 while considering the effects of both transmission and propagation over each segment, as well as processing delays at the routers.



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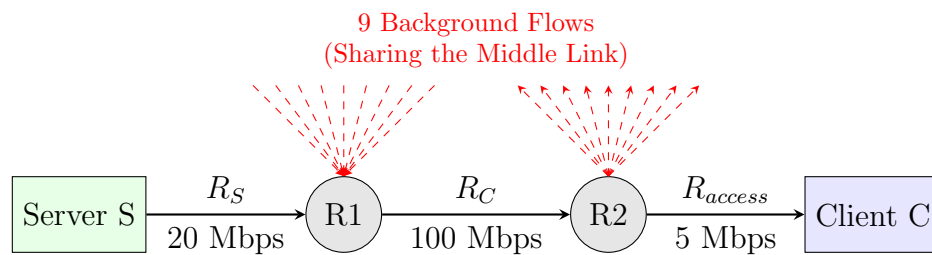
Quiz 1: Computer Networks and the Internet (Chapter 1)

Name: _____ Roll No: _____ Section: BSE-6B1 (Spring 2026)

1. (5 points) Throughput and Bottleneck Analysis

Consider the file transfer scenario shown below. Server S sends a file to Client C through two intermediate routers $R1$ and $R2$.

- Link $S \rightarrow R1$: Rate $R_S = 20$ Mbps.
- Link $R1 \rightarrow R2$: Shared link, Rate $R_C = 100$ Mbps. This link is shared by 9 other background traffic flows (assume all are working at full capacity).
- Link $R2 \rightarrow C$: Rate $R_{access} = 5$ Mbps.
- The 9 background flows consume exactly 10 Mbps each of the middle link ($R1 \rightarrow R2$).

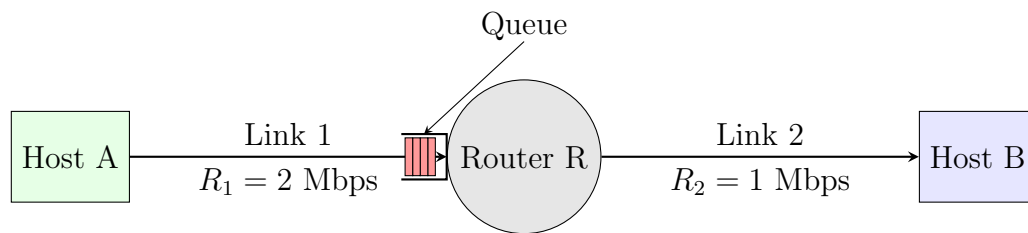


Calculate the throughput for the file transfer from S to C . Show the available bandwidth for each link and identify the bottleneck link.

2. (10 points) Nodal Delays and Queuing Dynamics

Consider a packet-switched network where Host A sends a burst of $N = 10$ packets back-to-back to Host B via a single Router R.

- **Link 1 (A $\rightarrow$ R):** Transmission Rate $R_1 = 2$ Mbps. Propagation Delay $d_{prop1} = 10$ ms.
- **Link 2 (R $\rightarrow$ B):** Transmission Rate $R_2 = 1$ Mbps. Propagation Delay $d_{prop2} = 10$ ms.
- **Packet Size:** All packets have a length $L = 2000$ bits.
- **Processing Delay:** Negligible at Router R ($d_{proc} \approx 0$).
- **Queuing:** Router R has a buffer. Since $R_1 > R_2$, packets arrive at R faster than they can be transmitted to B, causing a queue to build up.



- (a) (4 points) Calculate the **Transmission Delay** per packet for Link 1 (d_{trans1}) and Link 2 (d_{trans2}). Notice that a queue builds up at Router R.
- (b) (6 points) Calculate the **Queuing Delay** experienced specifically by the **10th packet** at Router R.

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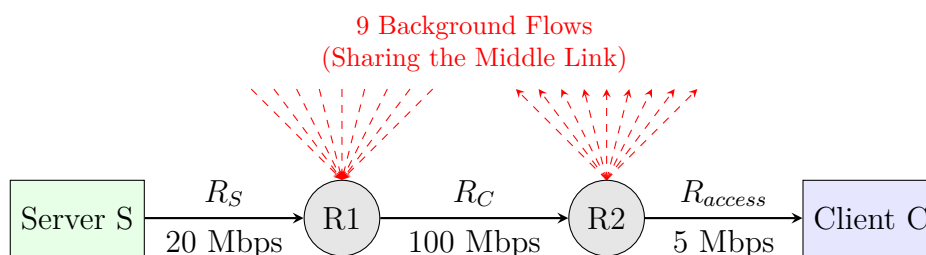
Quiz 1: Computer Networks and the Internet (Chapter 1)

Name: _____ Roll No: _____ Section: BSE-6B1 (Spring 2026)

1. (5 points) Throughput and Bottleneck Analysis

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- Link $R2 \rightarrow C$: Rate $R_{access} = 5$ Mbps.
- The 9 background flows consume exactly 10 Mbps each of the middle link ($R1 \rightarrow R2$).



Calculate the throughput for the file transfer from S to C . Show the available bandwidth for each link and identify the bottleneck link.

Solution: The end-to-end throughput is determined by the bottleneck link, which is the link with the minimum available bandwidth along the path.

$$\text{Throughput} = \min(R_S, R_C, R_{access})$$

- **Link $S \rightarrow R1$ (R_S):** Capacity is 20 Mbps. No background traffic.
Available = 20 Mbps.
- **Link $R1 \rightarrow R2$ (R_C):** Total Capacity is 100 Mbps.
Background traffic = 9 flows $\times$ 10 Mbps/flow = 90 Mbps used.
 $R_C = 100 - 90 = 10$ Mbps available for our flow.
- **Link $R2 \rightarrow C$ (R_{access}):** Capacity is 5 Mbps.
Available = 5 Mbps.

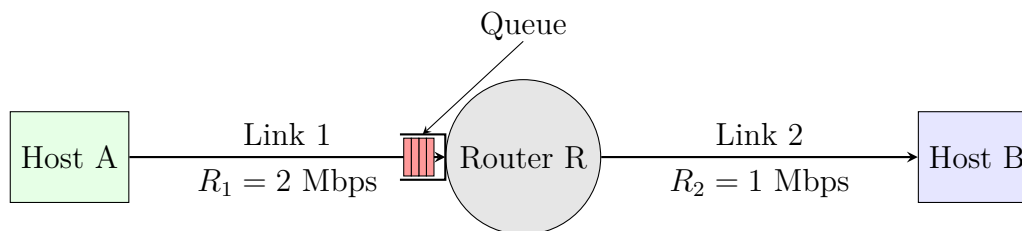
$$\text{Throughput} = \min(20, 10, 5) = \mathbf{5 \text{ Mbps.}}$$

The bottleneck is the client's access link (R_{access}).

2. (10 points) Nodal Delays and Queuing Dynamics

Consider a packet-switched network where Host A sends a burst of $N = 10$ packets back-to-back to Host B via a single Router R.

- **Link 1 (A → R):** Transmission Rate $R_1 = 2$ Mbps. Propagation Delay $d_{prop1} = 10$ ms.
- **Link 2 (R → B):** Transmission Rate $R_2 = 1$ Mbps. Propagation Delay $d_{prop2} = 10$ ms.
- **Packet Size:** All packets have a length $L = 2000$ bits.
- **Processing Delay:** Negligible at Router R ($d_{proc} \approx 0$).
- **Queuing:** Router R has a buffer. Since $R_1 > R_2$, packets arrive at R faster than they can be transmitted to B, causing a queue to build up.



- (a) (4 points) Calculate the **Transmission Delay** per packet for Link 1 (d_{trans1}) and Link 2 (d_{trans2}). Notice that a queue builds up at Router R.

Solution:

$$d_{trans1} = \frac{L}{R_1} = \frac{2000}{2 \times 10^6} = 0.001 \text{ s} = \mathbf{1 \text{ ms}}$$

$$d_{trans2} = \frac{L}{R_2} = \frac{2000}{1 \times 10^6} = 0.002 \text{ s} = \mathbf{2 \text{ ms}}$$

- (b) (6 points) Calculate the **Queuing Delay** experienced specifically by the **10th packet** at Router R.

Solution: Queuing delay is the time a packet waits in the buffer before it starts being transmitted.

$$d_{queue} = \text{Time Service Starts} - \text{Time Packet Arrives}$$

1. Arrival time of 10th packet at R:

Host A sends packets back-to-back. The 10th packet finishes transmission at A at $10 \times d_{trans1} = 10 \times 1 = 10$ ms. It travels over Link 1.

$$T_{arrival_10} = 10 \text{ ms} + d_{prop1} = 10 + 10 = \mathbf{20 \text{ ms}}$$

2. Start of service time for 10th packet at R:

The router transmits packets 1 through 9 before handling packet 10. Transmission of Pkt 1 starts at $t = 11$ ms. Time to transmit 9 packets = $9 \times d_{trans2} = 9 \times 2 = 18$ ms.

$$T_{start_10} = 11 \text{ ms} + 18 \text{ ms} = \mathbf{29 \text{ ms}}$$

3. Queuing Delay:

$$d_{queue_10} = T_{start_10} - T_{arrival_10} = 29 \text{ ms} - 20 \text{ ms} = \mathbf{9 \text{ ms}}$$

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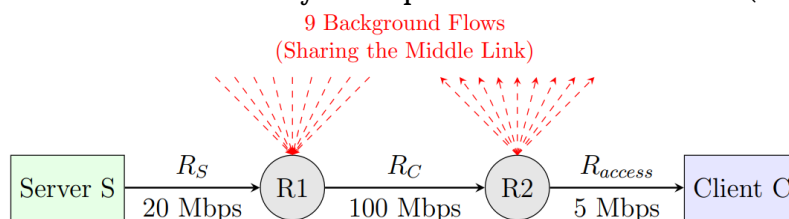
Section:

Q1. Carefully read the Question before attempting:

[5 marks]

Consider the file transfer scenario shown below. Server S sends a file to Client C through two intermediate routers R1 and R2.

- Link S → R1: Rate $R_S = 20$ Mbps.
- Link R1 → R2: Shared link, Rate $R_C = 100$ Mbps. This link is shared by 9 other background traffic flows (assume all are working at full capacity).
- Link R2 → C: Rate $R_{access} = 5$ Mbps.
- The 9 background flows consume exactly 10 Mbps each of the middle link (R1 → R2).



Calculate the throughput for the file transfer from S to C. Show the available bandwidth for each link and identify the bottleneck link.

Solution:

- Background consumption on middle link = $9 \times 10 \text{ Mbps} = 90 \text{ Mbps}$
- Available bandwidth on middle link = $100 \text{ Mbps} - 90 \text{ Mbps} = 10 \text{ Mbps}$

Available bandwidth on each link

- S → R1: 20 Mbps
- R1 → R2 (available): $100 - 90 = 10 \text{ Mbps}$
- R2 → C: 5 Mbps

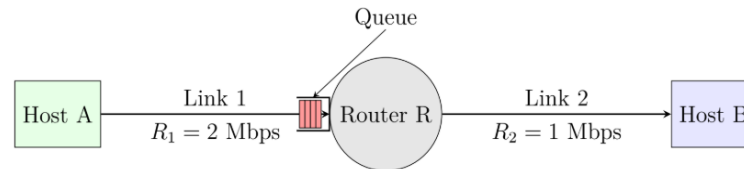
Throughput = minimum available along path = $\min(20, 10, 5) = 5 \text{ Mbps}$.

Bottleneck link: R2 → C (access link) at **5 Mbps**.

Q2. Carefully read the Question before attempting:**[10 marks]**

Consider a packet-switched network where Host A sends a packet to Host B via a single Router R.

- Link 1 (A $\rightarrow$ R): Transmission Rate $R_1 = 2$ Mbps. Propagation Delay $d_{\text{prop1}} = 10$ ms.
- Link 2 (R $\rightarrow$ B): Transmission Rate $R_2 = 1$ Mbps. Propagation Delay $d_{\text{prop2}} = 10$ ms.
- Packet Size: All packets have a length $L = 2000$ bits.
- Processing Delay: Negligible at Router R ($d_{\text{proc}} \approx 0$).
- Queuing: Router R has a buffer. Since $R_1 > R_2$, packets arriving at R faster than they can be transmitted to B, may cause a queue to build up.



1. Calculate the transmission delays at each Link?
2. What is the total end to end delay considering negligible queuing and processing delays (router must receive the whole packet before starting transmission)?

Solution:

Link 1 (A $\rightarrow$ R):

$$dt1 = 2000 \text{ bits} / 2,000,000 \text{ bits/s}$$

$$\text{Divide: } 2000 \div 2,000,000 = 0.001 \text{ s}$$

$$\text{Convert to ms: } 0.001 \text{ s} \times 1000 = 1 \text{ ms}$$

$$\text{So } dt1 = \mathbf{1 \text{ ms}}$$

Link 2 (R $\rightarrow$ B):

$$dt2 = 2000 \text{ bits} / 1,000,000 \text{ bits/s}$$

$$\text{Divide: } 2000 \div 1,000,000 = 0.002 \text{ s}$$

$$\text{Convert: } 0.002 \text{ s} \times 1000 = 2 \text{ ms}$$

$$\text{So } dt2 = \mathbf{2 \text{ ms}}$$

Total End-to-End Delay (store-and-forward, negligible queuing & processing):

$$\text{End-to-End Delay} = dt1 + d_{\text{prop1}} + dt2 + d_{\text{prop2}}$$

$$= 1 \text{ ms} + 10 \text{ ms} + 2 \text{ ms} + 10 \text{ ms}$$

$$= \mathbf{23 \text{ ms}}$$

SOLUTION QUIZ-01 BSE-6A

QUESTION-01

1. C
2. B
3. B
4. C
5. A

QUESTION-02

- a. Transmission Delay = Number of Cars x Service Time per Car = 20 cars x 2 sec/car = 40s
- b. Propagation Delay = Distance / Speed = 400 km / 10 kms⁻¹ = 40s
- c.

1. **Car 1 leaves Booth 1:** At t = 2 s.
2. **Requirement for Booth 2:** Service cannot start until the entire caravan (all 20 cars) has arrived at Booth 2.
3. **When does the last car (Car 20) arrive at Booth 2?**
4. Car 20 leaves Booth 1 at t = 40s (calculated in part a).
5. Car 20 travels for 40s (calculated in part b).
6. Car 20 arrives at Booth 2 at t = 40 + 40 = 80s.
7. **Service Start:** Booth 2 begins service at t = 80s.

Calculation:

Time from Car 1 leaving Booth 1 (t=2) to Car 1 entering service at Booth 2 (t=80).

Time = 80s - 2s = 78s

- d. No.
 - Booth 1 is busy servicing cars from t = 0 to t = 40.
 - Booth 2 cannot start servicing cars until the entire caravan arrives (at t=80). So Booth 2 works from t = 80 to t = 120.
- e. Yes. During the 40-second gap, the entire caravan is effectively "on the road" (propagating) or waiting in the queue at Booth 2, but no tollbooth is actively servicing a car.

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Q1. Encircle the correct option:

[CLO 1] [5 marks]

- Which of the following best describes the primary disadvantage of strict protocol layering (e.g., the OSI or TCP/IP model)?
 - It prevents the abstraction of complex hardware details.
 - It increases the complexity of error detection at the physical layer.
 - It may lead to redundant functionality (e.g., error recovery) being implemented in multiple layers.
 - It makes it impossible to change the implementation of one layer without affecting the others.
- Time needed to perform an integrity check, lookup packet information in a local table and move the packet from an input link to an output link in a router.
 - Queueing delay
 - Processing delay
 - Transmission delay
 - Propagation delay
- A link has a high bandwidth-delay product. This implies:
 - The link is physically short but very fast.
 - A large volume of data can be in flight before the first bit is acknowledged.
 - The propagation delay is negligible.
 - Retransmissions are inexpensive.
- If the Traffic Intensity is close to 1, what is the expected behavior of the queueing delay?
 - It approaches zero.
 - It becomes constant.
 - It grows exponentially/asymptotically toward infinity.
 - It is equal to the propagation delay.
- A _____ is formed at the transport layer by encapsulating an application layer message with transport layer headers.
 - Segment
 - Datagram
 - Message
 - Frame

Q2: Suppose a caravan has 20 cars, and the tollbooth services a car at a rate of one car per 2 seconds. Once serviced, a car proceeds to the next toll booth, which is 400 kilometers away at a rate of 10 kilometers per second. The entire caravan must be queued at a tollbooth before the first car in the caravan can be serviced (i.e., pay its toll and begin driving towards the next tollbooth). Please answer the following questions: **[2+2+4+1+1 = 10 Marks] [CLO 1]**

Hint: This is an example of Store-and-Forward packet switching.

a. How long does it take for the entire caravan to receive service at the tollbooth (that is the time from when the first car enters service until the last car leaves the tollbooth)?

b. Once the first car leaves the tollbooth, how long does it take until it arrives at the next tollbooth?

c. Once the first car leaves the tollbooth, how long does it take until it enters service at the next tollbooth?

d. Are there ever two cars in service at the same time, one at the first tollbooth and one at the second tollbooth? And Why?

e. Are there ever zero cars in service at the same time, i.e., the caravan of cars has finished at the first tollbooth but not yet arrived at the second tollbooth? And Why?

SOLUTION QUIZ-01 BSE-6B

QUESTION-01

1. B
2. C
3. A
4. D
5. A

QUESTION-02

Transmission Delay = $L/R = 5000 / 1000000 = 0.005\text{s} = 5\text{ms}$

Packet 1

1. **Host A sends Packet 1:** Ends at 5 ms.
2. **Propagates to Switch:** Arrives at switch at $5\text{ ms} + 0.02\text{ ms} = 5.02\text{ ms}$.
3. **Switch Processing:** $5.02\text{ ms} + 0.035\text{ ms} = 5.055\text{ ms}$.
4. **Switch forwards Packet 1:** Starts at 5.055 ms. Takes 5 ms to transmit. Ends at 10.055 ms.
5. **Propagates to Host B:** Arrives at destination at $10.055\text{ ms} + 0.02\text{ ms} = 10.075\text{ ms}$.

Packet 2 (Pipelining)

Host A sends Packet 2 immediately after Packet 1.

1. **Host A sends Packet 2:** Starts at 5 ms, ends at 10 ms.
2. **Propagates to Switch:** Arrives at switch at $10\text{ ms} + 0.02\text{ ms} = 10.02\text{ ms}$.
3. **Switch Processing:** $10.02\text{ ms} + 0.035\text{ ms} = 10.055\text{ ms}$.
4. **Switch forwards Packet 2:**
 - The switch finished sending Packet 1 at exactly 10.055 ms (from Step 2).
 - The switch is free, and Packet 2 is processed and ready at exactly 10.055 ms.
 - Transmission starts at 10.055 ms and takes 5 ms.
 - Transmission ends at 15.055 ms.
5. **Propagates to Host B:** $15.055\text{ ms} + 0.02\text{ ms} = 15.075\text{ ms}$.

Final Answer: 15.075 ms.

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Q1. Encircle the correct option:

[5 marks] [CLO 1]

1. What does it mean for a network to achieve a "high utilization"?
 - a. The network maintains low latency.
 - b. The network fully uses available bandwidth.
 - c. Packets are evenly distributed across all routes.
 - d. The network ensures no packet loss occurs.
2. Nodal processing delay involves all of the following except:
 - a. Checking for bit errors.
 - b. Determining the output link.
 - c. Waiting for the link to become idle.
 - d. Examining the packet header.
3. The Bandwidth-Delay Product (BDP) effectively measures:
 - a. Maximum number of bits present on the wire at any instant.
 - b. The total time it takes for a packet to traverse the link.
 - c. The speed at which bits are pushed onto the medium.
 - d. The queuing delay experienced by the last bit.
4. Consider a path with three links with rates $R_1 = 200$ kbps, $R_2 = 5$ Mbps, and $R_3 = 1$ Mbps. If a server transmits a large file to a client over this path, what is the theoretical maximum throughput?
 - a. 5 Mbps
 - b. 2.06 Mbps (Average)
 - c. 1 Mbps
 - d. 200 kbps
5. A _____ is formed at the transport layer by encapsulating an application layer message with transport layer headers.
 - a. Segment
 - b. Datagram
 - c. Message
 - d. Frame

Q2: Two hosts are connected via a packet switch with 10^6 bits per second links. The packet switch implements store-and-forward scheme, and each link has a propagation delay of 20 microseconds. The switch begins forwarding a packet 35 microseconds (time incurred on processing) after it receives that packet. If 10,000 bits of data are to be transmitted between the two hosts using a packet size of 5,000 bits, ignoring any queuing delays, please determine the time required to receive 10,000 bits at the destination. **[10 Marks] [CLO 1]**

